

Antiaromaticity Relief-Mediated Intramolecular Charge Transfer Enables Efficient and Stable Radical Emission in Polar Solvents

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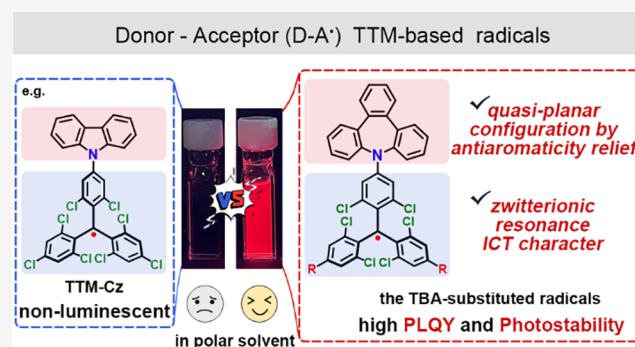


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Supporting Information

ABSTRACT: Triphenylmethyl (trityl) radicals have shown potential for use in organic optoelectronic and spintronic applications. However, due to the poor luminescence of alternant symmetry hydrocarbons, the design of practical trityl structures has been limited to donor-radical intramolecular charge-transfer systems to break the alternant symmetry, which results in poor emission and photostability in polar solvents. Here, we synthesized a series of donor–acceptor (D–A[•]) neutral radicals based on the tris(2,4,6-trichlorophenyl)methyl (TTM) radical moiety by introducing the 9H-tribenzo[*b,d,f*]azepine (TBA) ring with 20 π electrons as a donor group. In contrast to other TTM-based D–A[•] radicals, these TBA radicals exhibit unique and superior properties for luminescent efficiency (56% in acetonitrile) and photostability ($t_{1/2} = 4.3 \times 10^5$ s, with estimated half-lives reaching up to several days) in polar solvents. Transient absorption (TA) spectra reveal the occurrence of an efficient transformation from the Franck–Condon (FC) state to the zwitterionic resonance intramolecular charge-transfer (ICT) state with a time constant of a few picoseconds. Due to the zwitterionic resonance ICT character, nonradiative decay channels, which are dominated by high-frequency modes (over 1000 cm^{-1}) in TBA radicals, can be efficiently suppressed. The present study highlights the promotion of antiaromaticity relief in luminescent radicals and offers a valuable viewpoint for understanding the ICT excited-state dynamics of D–A-type dyads in polar solvents.



INTRODUCTION

Organic radicals¹ exhibit remarkable magnetic^{2–4} and electronic properties^{5–7} related to the presence of unpaired electrons. Luminescent neutral radicals^{8,9} featuring an emissive doublet excited state (D_1), particularly the air-stable tris(2,4,6-trichlorophenyl)methyl (TTM) derivatives, have recently emerged as promising candidates for photonic^{10–13} and optoelectronic,^{14,15} owing to the absence of energetically low-lying dark states in these systems. However, conventional TTM radicals generally suffer from a low luminescence efficiency and limited photostability. These shortcomings originate from their intrinsically weak extinction coefficient for the D_0 – D_1 transition in all solvents. More fundamentally, the alternant symmetry of the TTM core gives rise to degenerate orbital excitations of equal magnitude but opposite dipole moments, leading to vanishing oscillator strength for the lowest-energy D_0 – D_1 excitation (Figure 1a).^{16,17}

To circumvent these intrinsic limitations, several molecular design strategies have been proposed to “brighten” radical emitters. For example, as illustrated in Figure 1b, breaking the alternant symmetry enables radicals such as TTM-Cz to transform the formally forbidden D_1 state into a charge-transfer (CT) D_1 state, characterized by an electron transition from the donor HOMO (the highest doubly occupied molecular orbital) to the singly occupied molecular orbital (SOMO) of

the TTM core.¹⁶ Compared with a localized excitation, the CT excited state promotes more efficient luminescence in TTM radicals. Nevertheless, such donor–acceptor (D–A[•]) radicals exhibit pronounced sensitivity to solvent polarity, often resulting in unstable emission and diminished efficiency in polar environments—an issue particularly critical for applications in electroluminescence, bioimaging, and therapeutic technologies. This highlights a central challenge: the development of new molecular design principles that can simultaneously ensure high stability and efficient luminescence in diverse polar surroundings, while also provides a new viewpoint for understanding the CT-driven excited-state dynamics of D–A[•] radical systems.^{18,19}

Donor moieties with antiaromatic electronic structures,^{20,21} in accordance with formal Hückel’s $4n\pi$ rule, enable configuration mixing with low-energy zwitterionic states, thereby alleviating antiaromatic destabilization of the sub-

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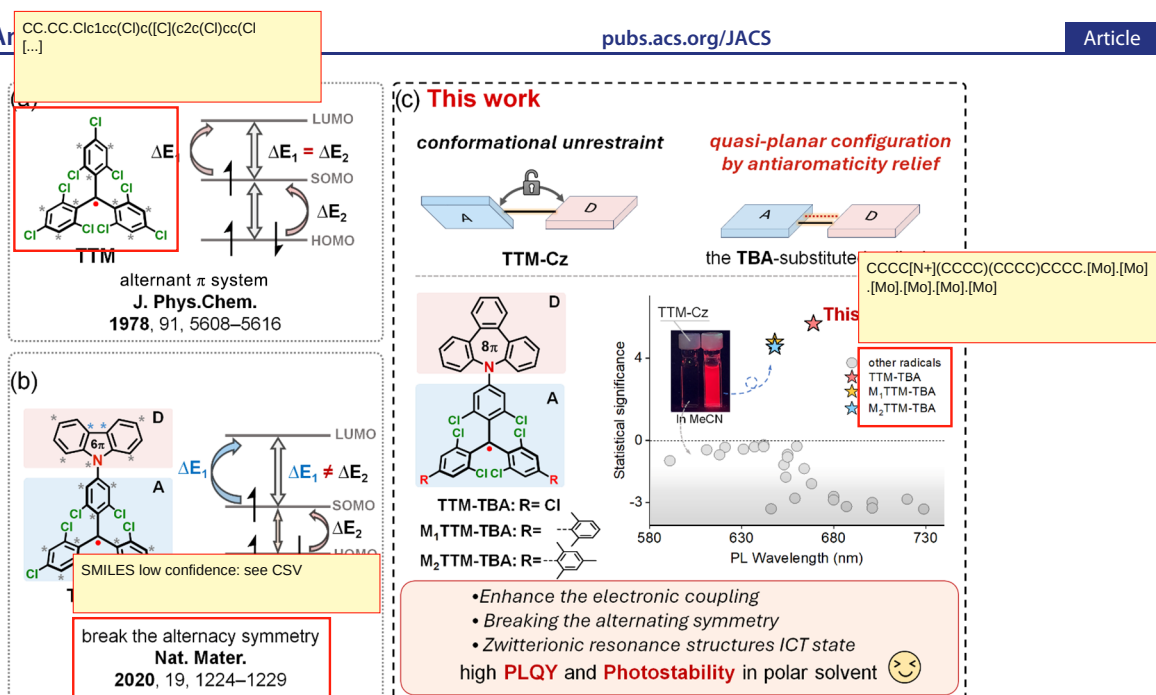


Figure 1. (a) Schematic diagram of the frontier molecular orbitals of TTM, with the degeneracy of HOMO-SOMO and SOMO-LUMO (the lowest doubly unoccupied molecular orbital). (b) TTM functionalized to a nonalternant group. (c) Different from other TTM-based D–A[•] radicals with conformational unrestraint (e.g., TTM-Cz), the TBA-substituted radicals with quasi-planar D–A[•] conformations by antiaromaticity relief. Illustration of the chemical structure and properties of the TBA-substituted radicals. Statistical significance represents the significant difference in the photoluminescence quantum yield (PLQY) of reported TTM-based D–A[•] radicals related to solvent polarity. The details are described in Table S1 (Supporting Information).

stituents.^{22–25} We envisioned that this effect could provide an effective strategy to mitigate the detrimental influence of solvent polarity on the stability and luminescence efficiency of TTM-based D–A[•] radicals. Herein, we report the synthesis and photophysical characterization of a series of TTM radicals functionalized with 9H-tribenzo[*b,d,f*]azepine (TBA) as a formally Hückel antiaromatic donor unit (Figure 1c). Through combined spectroscopic and computational investigations, we elucidate the mechanism by which antiaromaticity relief induces the excited-state dynamics of these D–A[•] systems. In sharp contrast to TTM-Cz with the aromatic donor group, our investigations revealed that the TBA-substituted TTM radicals have shorter C–N bond lengths and stronger electronic coupling between donor and acceptor units. Remarkably, owing to the antiaromaticity relief effect, the TBA-substituted radicals exhibit high photostability and luminescence quantum yield, even under increasingly polar environments. Our findings establish antiaromaticity relief as an efficient design principle for stable radical emitters, achieving high near-infrared (NIR) luminescence efficiency across diverse polar environments.

RESULTS AND DISCUSSION

Synthesis and Characterization

The synthetic routes of TTM-TBA, M₁TTM-TBA, M₂TTM-TBA, and reference TTM-Cz are depicted in Scheme S1. First, the TTM-TBA radical was obtained via a Buchwald-Hartwig coupling reaction between TTM and 9H-tribenzo[*b,d,f*]azepine (TBA), followed by radical conversion through dehydrogenation and one-electron oxidation. Then, the M₁TTM-TBA and M₂TTM-TBA radicals were prepared by a Suzuki-Miyaura coupling reaction between TTM-TBA and the respective arylboronic acid, following the same radicalization process. We found that only Buchwald-Hartwig coupling

reactions with the TTM radical rather than the precursor HTTPM proceeded in moderate to good yields (45–60%). The target radicals were characterized via MALDI-TOF mass spectrometry and electron paramagnetic resonance (EPR). Detailed synthetic procedures and structural characterization are described in the Supporting Information Figures S1–S11. Furthermore, density functional theory (DFT) calculations (Figure S12) confirmed that the unpaired electrons for these radicals were mainly located on the TTM unit. The electrochemical characteristics of the radicals were investigated via cyclic voltammetry measurements (Figures S13–S16 and Table S2).

Structural Properties

In order to compare the differences in properties of TTM-based D–A[•] radicals, the single-crystal structures of TTM-TBA, M₁TTM-TBA, M₂TTM-TBA, and reference TTM-Cz with an aromatic donor (carbazole) were determined by single-crystal X-ray diffraction (XRD) analyses. Detailed atomic coordinates and bond lengths are summarized in Tables S3–S6 and Figures S17–S20. As shown in Figures 2 and S17–S20, all sp² hybridized radical carbon atoms (C1) lie in a plane with three adjacent phenyl carbons, and the bond angles centered at radical carbon are all close to 120°. Interestingly, TTM-TBA and its derivatives exhibit quasi-planar donor–acceptor conformations, unlike the twisted D–A[•] structure of TTM-Cz, which has a large dihedral angle of 83.2° between the carbazole and the attached phenyl linker (Table S7). As for TTM-TBA, the dihedral angle (θ_2) between the phenyl linker and adjacent donor segments is just 0.4°. These comparative results indicate that the p_z orbitals of nitrogen atoms and the π orbitals of the bridged phenyl ring tend to be more parallel in the TBA-substituted radical, while it tends to be almost perpendicular in TTM-Cz. Besides, TTM-TBA, M₁TTM-TBA, and M₂TTM-TBA show shorter C5–N1 bond lengths

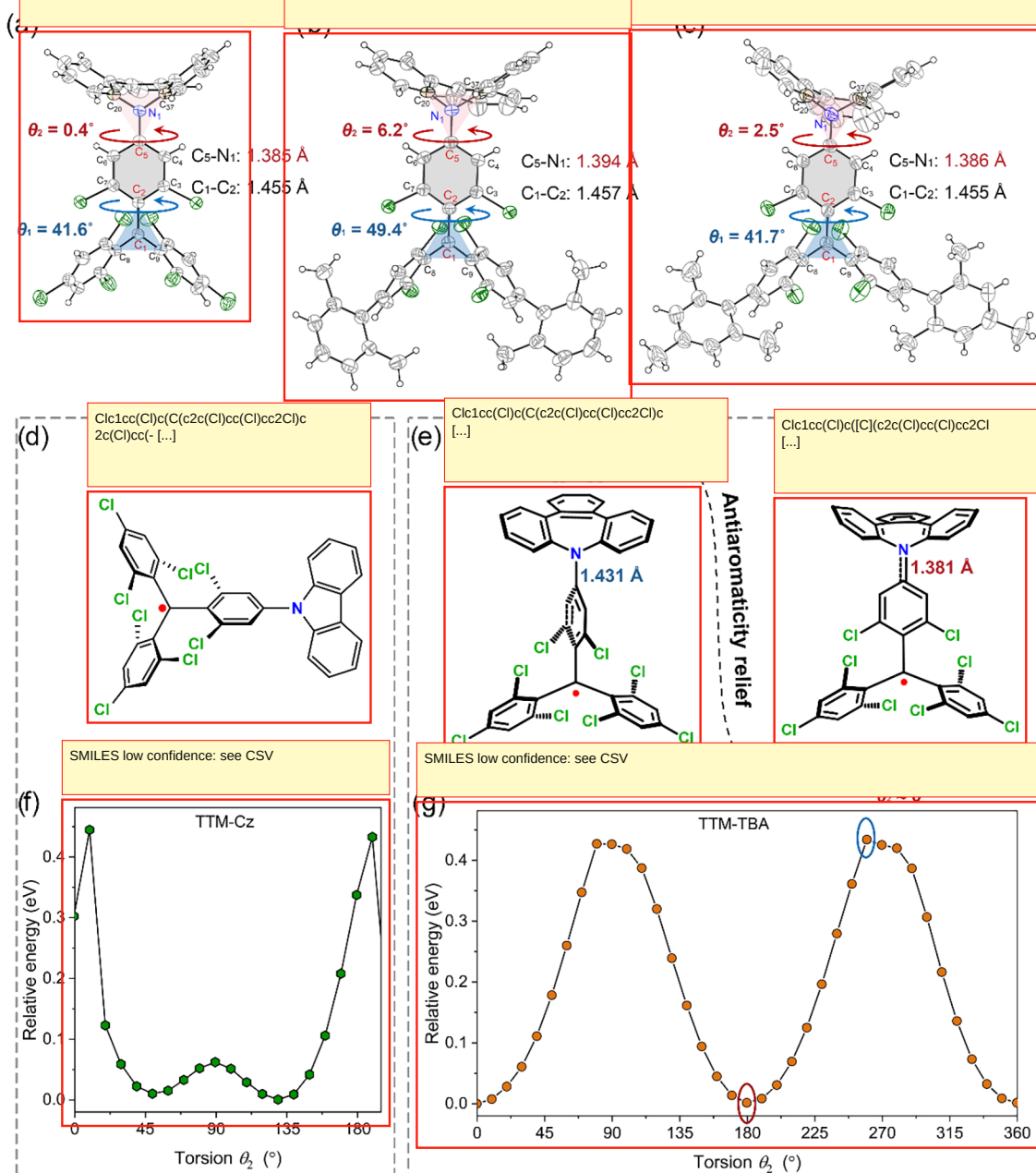


Figure 2. Crystal structures of (a) TTM-TBA, (b) M₁ TTM-TBA, and (c) M₂ TTM-TBA. (d) Illustration of the twisted D-A* configuration for TTM-Cz in the ground state and (f) energy of the ground state (relative to the lowest energy of the D₀ state) for TTM-Cz in cyclohexane as a function of the torsion (θ_2) between TTM and carbazole. (e) Illustration of the quasi-planar donor-acceptor conformation via antiaromaticity relief and (g) energy of the ground state (relative to energy of the ground state in P conformer) for TTM-TBA in acetonitrile.

(1.385–1.394 Å) than TTM-Cz (1.415 Å), as shown in Figure 2a–c. The shortened C–N bond lengths suggest the presence of some double-bond character. Furthermore, the IR spectra of all the TBA radicals feature the intense bands at around 1580 cm^{−1}, which is due to the C=N stretching vibration²⁶ (Figure S21). However, TTM-Cz exhibited normal C–N single bond ($\nu_{\text{C–N}} \approx 1463$ cm^{−1}) vibrational characteristics. Notably, the deviation from planarity in 9H-tribenzo[*b,d,f*]azepine (TBA) is due to benzene substituents. Although the TBA groups are formally antiaromatic (Figures S22–S25 and Table S8), the TBA radicals are best described as a nonaromatic system due to the sizable displacement from planarity.

Quantum-Chemical Modeling

To understand how nitrogen-containing fused ring donor moieties influence the ground state structure of TTM-based D-A* radicals, energies of the ground state for these radicals in acetonitrile were calculated at the UB3LYP/6–31G(d,p) level as a function of the torsion (θ_2) between the TTM substituent (Figures S26–S29). Different from TTM-Cz (Figure 2d), the quasi-planar configuration ($\theta_2 \approx 0^\circ$) for the TBA radicals is the most stable in the ground state, while its configuration has the highest energy at a torsion angle (θ_2) of 90° . We then further probed the influence of stereoelectronic interactions and steric effect on structure by setting the D-A* dihedral angle to 0° and 90° , respectively, to attain two geometry-optimized ground state conformations for the TBA radicals, the P conformer

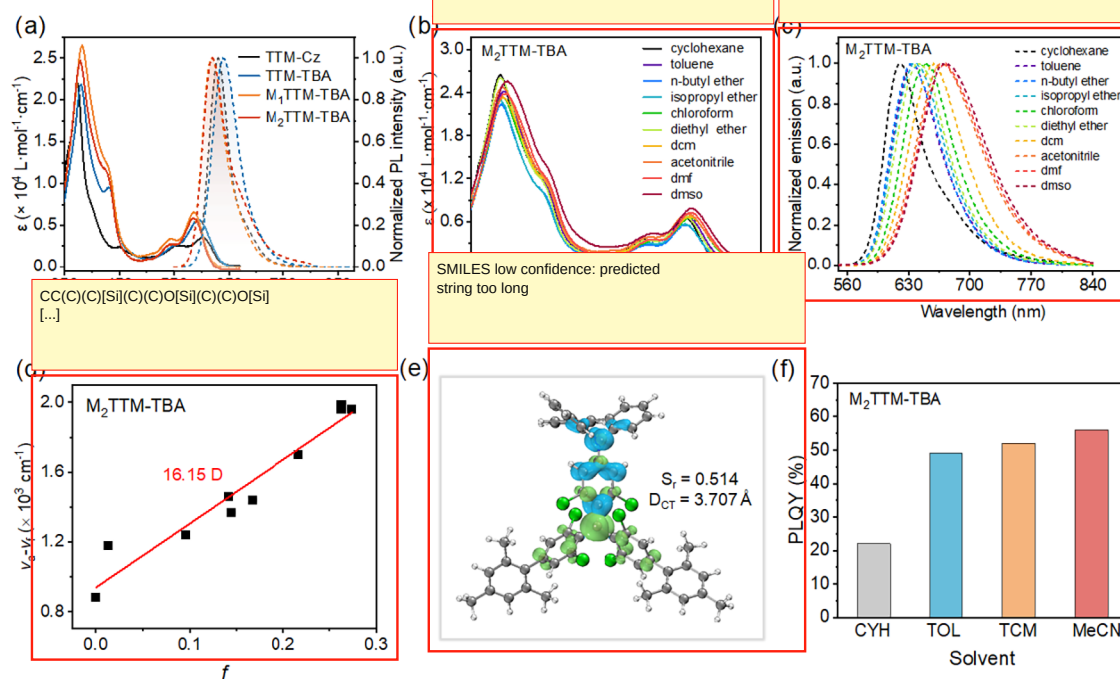


Figure 3. (a) Absorption and emission spectra of the four radicals in cyclohexane. The solvent-dependent (b) absorption and (c) emission spectra of M₂TTM-TBA. (dcm: dichloromethane, dmf: *N,N*-dimethylformamide, dms: dimethyl sulfoxide). (d) Lippert-Mataga fittings of the solvent-dependent spectroscopic data in M₂TTM-TBA. (e) Electron and hole distributions of the D₁ state of M₂TTM-TBA in chloroform. (f) Variation trend of PLQY of M₂TTM-TBA in solvents of different polarities (cyclohexane: CYH, toluene: TOL, chloroform: TCM, and acetonitrile: MeCN).

Table 1. Photophysical Parameters and Calculated the Electronic Coupling between the Donor and Acceptor Moieties (H_{ab}), Oscillator Strengths (f), Transition Dipole Moment ($\Delta\mu$) of the D₁ States of M₂TTM-TBA

M ₂ TTM-TBA	$\lambda_{PL}^{a,b}$	FWHM ^{a,c}	$\Phi_{PL}^{a,d}$	$\Delta\mu^{[t-dft]}$	f	$\Delta\mu^{[t-exp]}$	$\tau_{PL}^{a,e}$	$k_r^{a,f}$	$k_{nr}^{a,g}$	H_{ab}^h
CYH	618/2.01	50	22	4.64	0.1790	4.72	30.7	0.72	2.54	2176/270
TOL	632/1.96	64	49	4.75	0.1865	4.82	29.7	1.65	1.72	2220/275
TCM	648/1.91	73	52	5.16	0.2157	5.21	30.1	1.73	1.59	2313/287
MeCN	667/1.86	87	56	5.67	0.2536	5.64	29.8	1.88	1.48	2500/310

^aConcentration 10⁻⁵ M. ^bPhotoluminescence peak wavelength (nm/eV). ^cPhotoluminescence full width at half-maxima (nm). ^dPhotoluminescence quantum efficiency (%). ^ePhotoluminescence lifetime (ns). ^fRadiative decay rate (×10⁷ s⁻¹). ^gNonradiative decay rate (×10⁷ s⁻¹). [t-dft] calculated with DFT ΔE value (Debye). [t-exp] calculated with experiment ΔE value (Debye). ^hthe electronic coupling (H_{ab}) of D₁ state (cm⁻¹/meV).

(initial θ_2 set at 0°) and the V conformer (initial θ_2 set at 90°). The energies of all V conformers are ~0.4 eV higher than those of the P conformers (Table S9), and the C–N bond length (1.431–1.432 Å) is longer than that of the P configuration (1.381–1.383 Å), as shown in Figures 2e and S30–S31. This indicates that the quasi-planar donor–acceptor P conformer, with a shorter C–N bond length in accordance with single-crystal XRD analysis results, is the most stable ground state configuration.

Interaction Region Indicator (IRI) analysis indicates that steric hindrance between TBA and the TTM substituents exhibited similar situations in both P and V conformers (Figure S32). This indicates that the steric hindrance might not be the main reason for the restrained ground state structures. It is worth noting that the P conformer has shorter C3–C4 and C6–C7 bond lengths for the bridged benzene ring than those of the V configuration. Thus, we compared the computed nucleus-independent chemical shift (NICS) values²⁷ for the P conformer (V conformer as a reference) to evaluate the changes in the aromaticity of the rings (Figures S33–S34). For the P conformer, the NICS(1)_{zz} values of the azepine rings²⁸ become less positive, indicating a loss of antiaroma-

ticity. At the same time, the NICS(1)_{zz} values of the attached phenyl ring in the TBA radicals become less negative, which is consistent with the change of the C=C bond length on its structure, suggesting a loss of aromaticity. According to previous research,²⁵ antiaromaticity relief is an approach to stabilize radicals where the radical center acts as an electron “sink” to relieve the electronic suffering of an attached formally antiaromatic substituent. Our research results further indicate that antiaromaticity relief, originating from the attachment between the TTM core and the formally antiaromatic TBA unit, can enhance the D–A[•] conjugation, which leads to the restrained ground state structure of the TBA radicals (Figures 2e and S35, Table S10).

Steady-State Photophysical Properties

As shown in Figure 3a,b, the UV–vis absorption spectra of all radicals in cyclohexane show the typical high-energy D₀ → D_n excitation band of TTM peaking in the UV spectral region, whereas the lowest-energy D₀ → D₁ excitation is observed at around 500–670 nm (≈1.85–2.48 eV). When going from TTM-Cz to TBA-substituted D–A[•] radicals, a stronger feature is observed at around 435 nm. It is well to notice that the

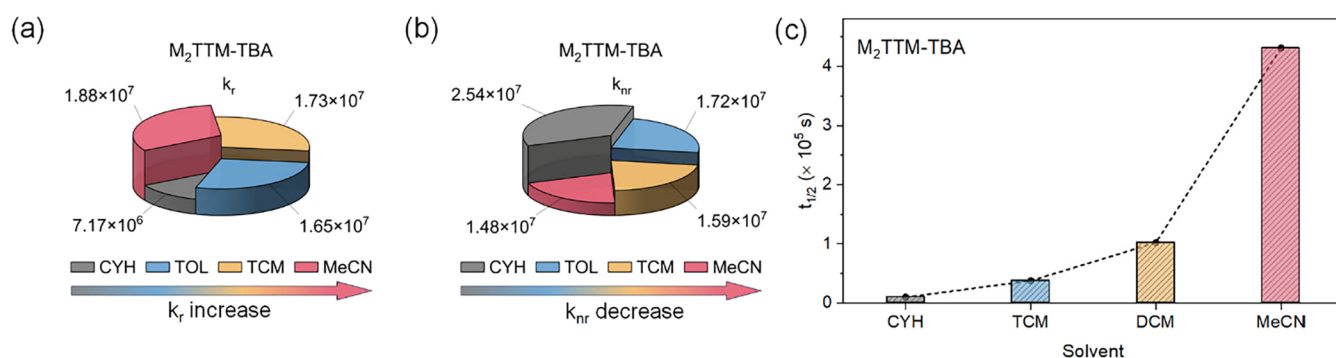


Figure 4. (a) Radiative decay rate constants (k_r) and (b) nonradiative decay rate constants (k_{nr}) of M_2 TTM-TBA in different solvents with variable polarity. (c) Photostability half-lives ($t_{1/2}$) of M_2 TTM-TBA in different solvents.

molar extinction coefficients (ϵ) of the lowest-energy excitation bands for TTM-TBA ($5700 \text{ M}^{-1} \text{ cm}^{-1}$), M_1 TTM-TBA ($5890 \text{ M}^{-1} \text{ cm}^{-1}$), and M_2 TTM-TBA ($6550 \text{ M}^{-1} \text{ cm}^{-1}$) are significantly higher than that of TTM-Cz ($3520 \text{ M}^{-1} \text{ cm}^{-1}$). The reason for this phenomenon can be attributed to two aspects. The TBA radicals introduce nonalternant hydrocarbon groups to break the alternant symmetry similar to TTM-Cz.¹⁶ On the other hand, **antiaromaticity relief** (Figure S35) can effectively enhance the electronic coupling (H_{ab}) of D_1 state according to eq S1 (Tables 1 and S11–S14), which originates from the p- π interaction between the donor nitrogen atom p_z orbital and the bridged benzene ring π orbital (Tables S15–S16).

The emission spectra of TTM-Cz and the TBA-based radicals in various solvents are shown in Figures 3c and S36–S38. The fluorescence spectrum of TTM-Cz (Figure S38) shows a strong solvatochromic effect (TTM-Cz: $\Delta\lambda = 59 \text{ nm}$) compared with that of the TBA radicals (TTM-TBA: $\Delta\lambda = 28 \text{ nm}$; M_1 TTM-TBA and M_2 TTM-TBA: $\Delta\lambda = 33 \text{ nm}$, from cyclohexane to chloroform). According to the Lippert-Mataga model, the D_1 excited states of TTM-TBA (16.14 D), M_1 TTM-TBA (15.87 D), and M_2 TTM-TBA (16.15 D) exhibit hybridized local and charge-transfer (HLCT) character,^{16,29} whereas TTM-Cz (21.82 D) shows charge-transfer (CT) characteristics in polar solvents (Figures 3d and S39). This assignment is confirmed by the electron hole distribution analysis shown in Figures 3e and S40. Interestingly, M_1 TTM-TBA and M_2 TTM-TBA radicals are luminescent in polar solvents, such as acetonitrile and dimethyl sulfoxide. Furthermore, their photoluminescence quantum yields (PLQYs) increase with increasing solvent polarity (Figures 3f and S41), a trend that sharply contrasts with previously reported D–A $\cdot$ luminescent radicals, which typically suffer fluorescence quenching in polar solvents.

To further investigate this abnormal solvatochromic effect, transient photoluminescence decay spectra of TTM-TBA, M_1 TTM-TBA, and M_2 TTM-TBA were measured in different solvents. With increasing solvent polarity, the decay lifetime of traditional D–A $\cdot$ type luminescent radicals such as TTM-Cz rapidly decreases.^{30,31} However, Figures S42–S44 exhibit anomalous photophysical properties, with fluorescence lifetimes of 21.7 ns (TTM-TBA), 10.8 ns (M_1 TTM-TBA), 10.8 ns (M_2 TTM-TBA) in cyclohexane and 29.6 ns (TTM-TBA), 19.3 ns (M_1 TTM-TBA), and 29.7 ns (M_2 TTM-TBA) in toluene, respectively. After that, their decay lifetime remains almost unchanged with increasing solvent polarity. According to eq S5, the values of radiative decay rate constants (k_r), and

nonradiative decay rate constants (k_{nr}) of the radicals were calculated in different solvents (Figures 4a,b, Tables 1 and S13–S14). **TTM-TBA exhibited increased radiative rates and decreased nonradiative rates from nonpolar solvent (cyclohexane) to medium polarity solvent (chloroform).** Especially, as shown in Figure 4a,b, M_2 TTM-TBA with two additional substituents still kept the trend in large polar solvents (acetonitrile). Moreover, we investigated the excited-state properties of the radicals by means of density functional theory (DFT) calculations performed at the UB3LPY/6–31G** level. The energies of several lowest five excited doublet (D_n) states are depicted in Figure S45. **Both the transition dipole moments ($\Delta\mu$) and oscillator strengths (f) of the D_1 excited state increase gradually as the solvent polarity increases** ($\Delta\mu$: $4.83 \rightarrow 5.59$ for TTM-TBA; $4.70 \rightarrow 5.64$ for M_1 TTM-TBA; $4.72 \rightarrow 5.64$ for M_2 TTM-TBA; f : $0.1840 \rightarrow 0.2449$ for TTM-TBA; $0.1775 \rightarrow 0.2537$ for M_1 TTM-TBA; $0.1790 \rightarrow 0.2536$ for M_2 TTM-TBA), according to the Einstein equation (eq S6), which is consistent with the trend of the increase in experiment calculated k_r . At the same time, the suppressed k_{nr} is related to the reduced reorganization energies (Table S17) of the radicals by means of the Marcus-Levich-Jortner approach (eq S8).

In addition, as shown in Figures 4c and S46–S49, continuous UV irradiation measurements ($\lambda_{\text{ex}} = 375 \text{ nm}$) revealed good photostability for TTM-TBA, M_1 TTM-TBA, and M_2 TTM-TBA in polar solvents. The half-life ($t_{1/2}$) values of $8.80 \times 10^5 \text{ s}$ in DCM and $4.31 \times 10^5 \text{ s}$ in MeCN were recorded for TTM-TBA and M_2 TTM-TBA, respectively. In contrast, TTM-TBA and M_2 TTM-TBA show inferior $t_{1/2}$ in cyclohexane (3.62×10^4 and $9.76 \times 10^3 \text{ s}$), which is opposite to the behavior of TTM-Cz. Notably, the emission of M_2 TTM-TBA in acetonitrile is stable, and it is the first D–A $\cdot$ -type luminescent radicals with good photostability in polar solvents, with estimated half-lives reaching several days.

TA Spectroscopy

It is of great significance to obtain further insight into the difference in the photophysical properties between TTM-Cz and the TBA radicals. **Considering that the structural and steady-state photophysical properties between M_1 TTM-TBA and M_2 TTM-TBA are quite similar, we performed femto-second transient absorption (fsTA) measurements of only TTM-Cz, TTM-TBA, and M_2 TTM-TBA to investigate their excited-state dynamics.** For all radicals, the pump pulse was resonant with a charge-transfer transition, which corresponds to a doublet excitation ($D_0 \rightarrow D_1$) from the Cz/TBA-centered highest doubly occupied molecular orbital (HOMO) to a

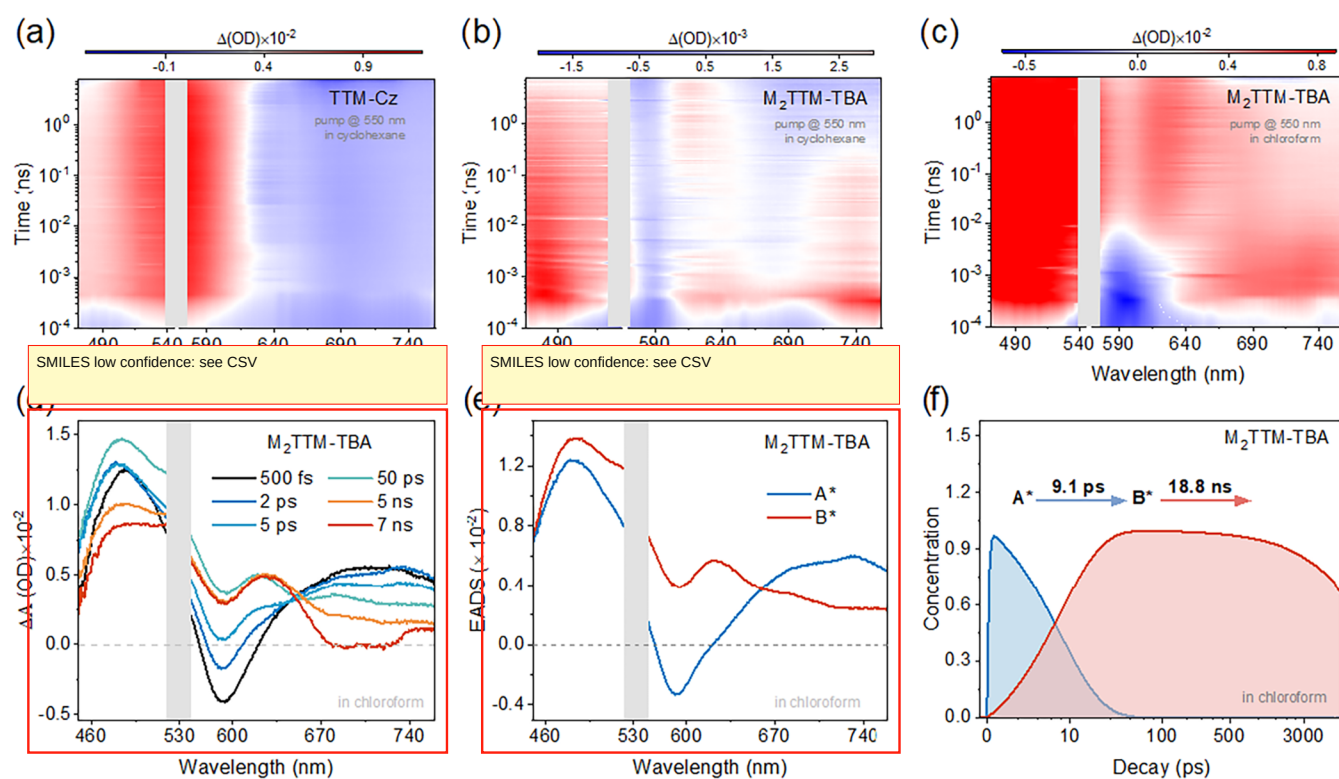


Figure 5. 3D fsTA spectra of (a) TTM-Cz and M₂TTM-TBA (b) in CYH and (c) TCM at 298 K. (d) fsTA spectra of M₂TTM-TBA at different time scales, (e) EADS spectra extracted from fsTA spectra, and (f) concentration profiles (Cp) of transient species after 550 nm excitation in TCM.

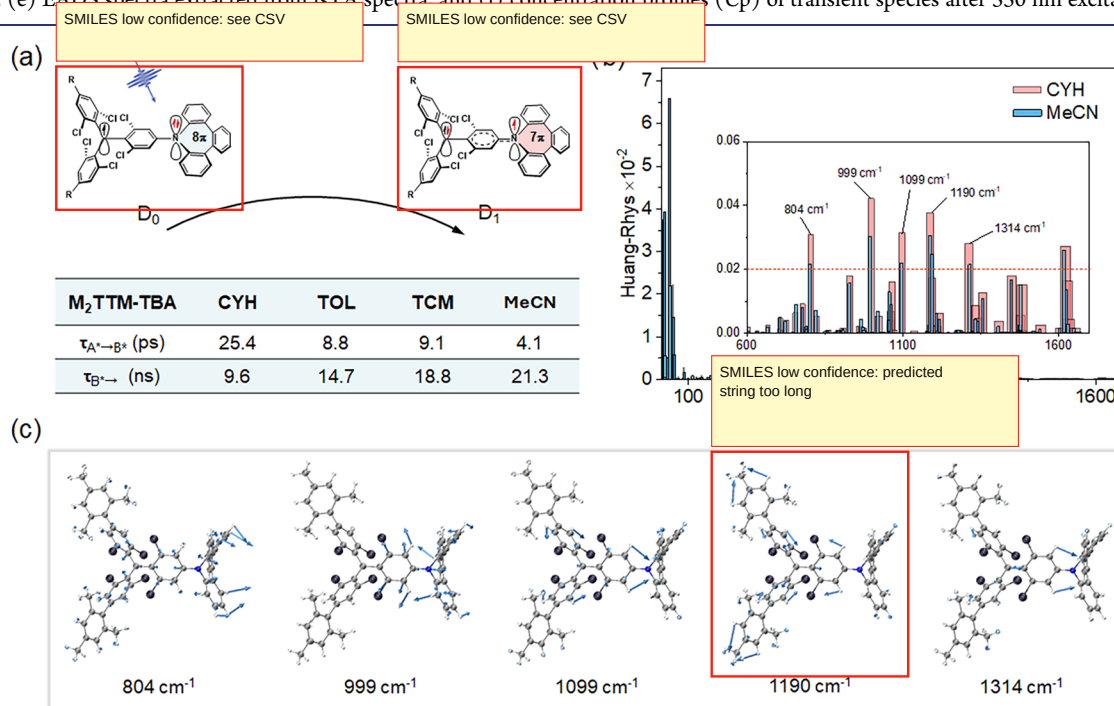


Figure 6. (a) Schematic diagram of intramolecular charge-transfer state dynamics in different solvents with variable polarity. (b) Theoretically calculated Huang–Rhys factor in cyclohexane and acetonitrile for D₀ → D₁ electronic transition and (c) vector displacement diagram of significantly different high-frequency breathing modes plotted on the optimized geometry of M₂TTM-TBA.

TTM-centered singly occupied molecular orbital (SOMO), as shown in Figures S51–S54.

The TA spectra of TTM-Cz in cyclohexane feature two prominent excited-state absorption (ESA) bands at ~570 and ~720 nm (Figure 5a). Qualitatively similar excited-state dynamics to those measured in cyclohexane were found in

all solvents (Figure S55). The strong ESA band at ~570 nm is attributed to the D₁ → D_n transition with predominant LE character, while the weak and broad ESA band at ~720 nm is associated with a CT-type transient, consistent with previous reports.³² A negative absorption band appearing after 4 ns between 600 and 750 nm is a combination of the ground state

bleach (GSB) and stimulated emission (SE). Notably, with the increase of solvent polarity, the lifetimes of two characteristic ESA bands from 500 to ~ 760 nm in the spectra of TTM-Cz decay rapidly from a few nanoseconds in cyclohexane to a few picoseconds in acetonitrile. This very short lifetime explains why the fluorescence of TTM-Cz in highly polar solvents cannot be detectable by stationary fluorescence spectroscopy. Comparatively, the TBA radicals have a much longer-lived fluorescence lifetime (20–30 ns) in different solvents, as determined from time-correlated single photon counting (TSCPC) measurements (Figures S42–S44, Tables 1 and S13–S14). Moreover, the fsTA spectra of the TBA radicals in various solvents (Figures 5b,c and S56–S57) exhibit spectacular changes in the first few picoseconds after excitation. At an early time, the spectrum of M_2 TTM-TBA in TCM consists of a negative GSB band at around 590 nm and two positive absorption bands at around 480 and 710 nm (Figure 5c), which can be assigned to the ESA of the D_1 state. During the first few picoseconds, the ESA band at 710 nm decays rapidly and a new one grows at around 625 nm. This change is associated with a clear isosbestic point around 650 nm, indicative of a precursor-successor relationship between the two states and species associated with these bands (Figure 5d).

To comprehensively investigate the contribution of transient species to spectra throughout the entire evolutionary process, a global target analysis³³ was conducted on fsTA data of TTM-TBA and M_2 TTM-TBA in different solvents, a two-step sequential model ($A^* \rightarrow B^* \rightarrow$) was implemented to reproduce the experimentally observed TA data together with the corresponding evolution-associated difference spectra (EADS), as shown in Figure 5e,f. EADS A^* is dominated by the D_1 state of M_2 TTM-TBA with absorption bands at 590 nm corresponding to steady-state absorption. Consequently, the A^* step corresponds mainly to the initially populated Franck–Condon (FC) state. EADS B^* is substantially different and can be seen from the disappearance of GSB characteristics at 590 nm and the emergence of the ESA characteristics at 625 nm associated with the 9H-tribenzo[*b,d,f*]azepine cation³⁴ (as shown in Figures S58–S59). Therefore, the $A^* \rightarrow B^*$ step, with a time constant of picoseconds scale, can be assigned to the antiaromaticity relief of the radical D_1 state to the zwitterionic resonance intramolecular charge-transfer (ZRICT) state.^{35,36} The relaxation of the latter state is clearly visible in the four solvents of different polarities. Interestingly, as shown in Figure 6a, the evolution time from A^* to B^* gradually accelerates in solvents of increasing polarity ($A^* \rightarrow B^*$: 25.4 ps in CYH, 8.8 ps in TOL, 9.1 ps in TCM, 4.1 ps in MeCN). Moreover, in contrast to TTM-Cz, the decay time of EADS B^* is gradually larger, indicating that EADS B^* can be stabilized by the solvent polarity.

Vibrational Analysis and Structural Relaxation

As shown in Figure 6a and Table S18, according to interfragment charge-transfer (IFCT) analysis, the D_1 state of TTM-TBA is an intramolecular charge-transfer state, with a significant increase in the dihedral θ_1 in Table S7 and C1–C2 bond length in Table S19. However, the change in dihedral θ_2 is not significant, and the double-bond characteristics of N1–C5 are enhanced. This leads to such a phenomenon that the TBA moiety is locked on the TTM moiety rather than rotating. As shown in Figure 6b, the four key high-frequency vibrational modes were identified for the $D_1 \rightarrow D_0$ transition of the TBA

radicals due to considerable Huang–Rhys (HR) factor, namely, 804, 999, 1099, and 1314 cm^{-1} , which is associated with C–N, C–C, and C=C stretching vibrations between the donor TBA moiety and TTM moiety (Figure 6c). According to previous research,^{37,38} the HR factors related to high-frequency modes are considered to promote fast nonradiative decay in organic molecules. The calculations show that for the four high-frequency modes, the vibrational coupling is significantly reduced by varying the polarity of solvents. Meanwhile, we found that the TBA radical modified with R groups such as mesityl effectively suppressed the stretching vibration of TTM skeleton at 1189 cm^{-1} (with increasing solvent polarity), which is significantly different from TTM-TBA (Figures S60–S61). This may explain why the M_1 TTM-TBA and M_2 TTM-TBA radical remains efficiently luminescent even in highly polar.

CONCLUSIONS

In summary, this work establishes antiaromaticity relief as an effective molecular design concept for constructing stable and efficient luminescent radicals in polar environments. As the antiaromaticity relief originating from the attachment between the TTM core and the formally antiaromatic TBA unit, all the TBA radicals show a quasi-planar donor–acceptor conformation. Thus, our results demonstrate that antiaromaticity relief can effectively enhance the D– A^* electronic coupling and stabilize the ICT excited state. These features enable trityl radicals to achieve bright emission with exceptional photostability, even in polar solvents, a long-standing challenge in radical photophysics.

Furthermore, TA spectra reveal that antiaromaticity relief accelerates the FC state $\rightarrow$ ZRICT state transformation, which provides a deeper understanding of excited-state dynamics in radical systems and points to new opportunities for molecular engineering beyond conventional aromatic donor design. Finally, we found that, due to the ZRICT character, nonradiative decay channels, which are dominated by high-frequency vibrations corresponding to the C–N, C–C, and C=C stretching vibrations in TBA radicals, can be efficiently suppressed.

Overall, this strategy offers a generalizable platform for designing high-performance radical emitters in polar environments suitable for optoelectronic, photonic, and quantum information applications, paving the way for next-generation radical-based functional materials.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/jacs.5c19217>.

Experimental details, including synthesis, characterization data, and DFT calculation methods (PDF)

Accession Codes

Deposition Numbers 2494986–2494987 and 2500468 contain the supplementary crystallographic data for this paper. These data can be obtained free of charge via the joint Cambridge Crystallographic Data Centre (CCDC) and Fachinformationszentrum Karlsruhe Access Structures service.

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All authors have given approval to the final version of the manuscript.

Notes

The authors declare no competing financial interest.

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